

Certified analytical bounds for hypersonic reachable sets

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Abstract

This is a proposal in nonlinear analysis and dynamical systems. The object is a singularly perturbed family of differential inclusions in which compactness fails along the time axis: as the parameter vanishes, the motion that carries the important state transitions occupies a vanishing time, so it disappears from the ordinary limits of the family while continuing to decide where the trajectories go. The remedy we propose is a *profile decomposition* — a low-dimensional *skeleton*, finitely many concentrating *bubbles*, and a *residual* bounded in a stated metric — and the question is whether that decomposition is a theorem about solution sets rather than a description of them.

We state the question as a hierarchy. *Theorem A*: every sequence of trajectories splits, along a subsequence, into a skeleton, finitely many rescaled profiles above any fixed size, and a remainder. *Theorem B*: the profiles induce a set-valued jump map. *Theorem C*: the solution set of the family is contained in that of the skeleton-and-jump system, thickened by an explicit ball. This document proves Theorem C, with every constant written down, for a model whose potential is asymmetric in the one way that produces concentration. It does not prove Theorem A, and it is worth being exact about the relation: the theorem proved here establishes no decomposition, and its proof uses none. What it does is twofold. It is the quantitative estimate any decomposition would have to reproduce, including the count of profiles above a given size and the amplification they compound — both of which a Lyapunov functional supplies here *a priori*, where the classical argument obtains them by contradiction and without a constant. And it shows that the concentration structure can be reached without the decomposition machinery at all, by an exact integral identity and a budget. Whether the machinery then buys anything the elementary route does not is the first question of the programme, and we think it is a real one.

Hypersonic atmospheric entry is the example that supplies the problem and the one against which the result is to be measured; it is not the subject. [Section 7](#) states the correspondence and what the full system adds. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is the solution set of a family of differential inclusions: every state reachable under every admissible control and every admissible selection from the field. Two things make it hard to enclose. The first is its dimension, which puts the exact methods out of reach. The second is the nature of the enclosure: a bound is worth having only if it comes with a proof, and the proof is to be a pair of functions whose superlevel set contains the solution set — exhibited and checked, rather than produced by a solver and inheriting its accuracy ([Section 6](#)). This document is narrower than the programme and prior to it: it takes the one analytical difficulty on which the programme turns, strips it of the application, and proves a theorem about it.

The difficulty is a singular limit of a kind that defeats the ordinary compactness arguments. The example supplies the parameter: for atmospheric entry it is the ratio $\varepsilon = H_s/R_e$ of the atmospheric scale height to the planetary radius, whose square root is the ratio of the period of the oscillation about equilibrium to the slow time of the flight. As $\varepsilon \rightarrow 0$ the motion separates into arcs on which the forcing is not felt and *passes* in

which it is felt strongly for a vanishing time. A pass becomes instantaneous and keeps its effect, so in the limit a trajectory is a sequence of arcs joined by jumps — and yet the natural limiting objects do not see the jumps, because they weigh an interval of time by its length and a pass has none. A solution set is a support and not an integral, so what has no mass still counts.

This is the structural phenomenon of concentration, and the language for it is a profile decomposition: a low-dimensional *skeleton*, which is what survives the limit; one *bubble* per pass, the pass rescaled to its own width, whose net effect on the surviving coordinates is a jump; and a *residual*, which is what the skeleton and the finitely many bubbles do not capture. In the theory of nonlinear dispersive equations such a decomposition is obtained by iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure: it says which pieces can occur and that nothing else can, by contradiction, and it produces no constant. The authors of the wave-equation theory are explicit that the uniform bound following from their decomposition carries a constant they cannot describe [2, Corollary 2].

Our proposal is that here the structure can be had *quantitatively*, because the family carries a Lyapunov functional. The dissipation along a trajectory is monotone and its total variation is bounded uniformly in the parameter, so the number of profiles above any fixed size is bounded in advance rather than extracted by a contradiction argument, and the same functional bounds the time the motion may spend concentrating and the accumulated growth of the comparison. The bound on the count is Eq. (3.14) and it is proved below; what is not proved is the decomposition it would organise. We therefore state the question as the hierarchy of the abstract — the decomposition, the jump map it induces, and the inclusion for solution sets — and this document proves the third for a model that keeps the concentration and discards everything else.

Remark 1.1 (What the small parameter is, and what it is not). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty: if the excursion in y lasts a time of order ε and the forcing is of order one, its contribution to z is of order ε , so in the limit it leaves no trace and ordinary averaging describes everything. What produces a jump is not a large forcing imposed by hand but an *asymmetric potential* together with a scaling that keeps the oscillation slow while the asymmetry sharpens. In the model of Section 2 the air supplies a wall that steepens without bound below the equilibrium while gravity is uniform above it, so a large oscillation rebounds in a short time and coasts for a long one; and because ε enters only as the *width of the band* in which the air is felt, a rebound occupies a vanishing fraction of its orbit while delivering an impulse of order one. Lemmas 3.4 and 3.5 are those two statements. Concentration is the asymmetry of the well read at the right scale; it is not an assumption about the size of a force.

1.1 What this document does not prove

Theorem 4.2 is Theorem C of the hierarchy, for one model. Three further statements belong to the programme and are not proved here; we name each with the role it plays, so that what is missing is visible rather than implied.

Theorem A: the decomposition itself. That every sequence of trajectories with $\varepsilon_n \rightarrow 0$ splits, along a subsequence, into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder. Theorem 4.2 compares a trajectory with a trajectory of the limit system and does not assert this; what the Lyapunov functional of Assumption 2.4 supplies is the bound on the number of profiles that the classical argument obtains by contradiction — it is Eq. (3.14), and it is already proved — so the decomposition is expected to follow quantitatively. Proving it is *objective 1*.

The limit of the family in the weak topology. The dynamics become a linear constraint when the family is embedded in a space of occupation measures, and the solution set is then the support of a terminal measure. Section 6 gives the definitions and the topology and states what the companion to Theorem 4.2 would assert at that level; proving it is *objective 4*.

The passage from the weak limit back to the support. A solution set is a support and not an integral. Granting the limit of the measures and the profiles it carries, that the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain; the proof below obtains the inclusion from trajectories directly and so does not cross it. Which route the general case should take is open, and it is the question we would most value an advisor's judgment on.

One further obligation is of a different kind and is deferred rather than open. A bound in closed form still has to be evaluated, and evaluation in floating point is where a guarantee is ordinarily lost; the setting

in which it is not is validated interval arithmetic, in which every operation is rounded outward so that the computed interval encloses the exact result [3, 4, Ch. 3]. That is a question about evaluating a bound rather than about proving one, and it returns as [objective 5](#).

1.2 Terminology

Two fields meet here and several words mean different things in each. We fix them once.

term	what it means here
<i>orbit</i>	One closed curve of Eqs. (2.1) to (2.2) in the (z, v) plane at <i>frozen</i> (c, θ) and zero damping: a single complete oscillation, in the sense of Lemma 2.8 . <i>Not</i> a trajectory about the planet — the motions here are sub-orbital and decelerating throughout — and <i>not</i> a piece of an actual trajectory, which closes no orbit. Orbits serve only as the comparison objects that define P_0 and D_0 in Lemma 3.4 ; the decomposition of an actual trajectory is Definition 3.6 .
<i>pass, dip</i>	A maximal interval of the actual trajectory on which $z \leq 0$ (Definition 3.6).
<i>arc</i>	A maximal interval on which $z \geq 0$.
<i>selection θ</i>	One member of the admissible family of fields, <i>fixed along a trajectory</i> and quantified over at the end (Remark 2.3); never a function of time in what is proved here.
<i>profile m_θ, density m_θ^ε</i>	The <i>profile</i> is the ε -free function of Assumption 2.1 , read in units of the band's own thickness; the <i>density</i> is the profile as the motion meets it, $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$. Hypotheses are placed on the profile, which is why they are uniform in ε ; the equations of motion carry the density.
<i>bubble</i>	A pass rescaled to its own width: a concentrating profile, whose net effect on the slow coordinates is a jump.
<i>skeleton</i>	What survives the limit: the arcs between passes, and the glide.
<i>residual</i>	The ball by which the inclusion is thickened.
<i>amplitude W</i>	The energy $\frac{1}{2}v^2 + \Phi_{c,\theta}(z)$ of the oscillation, and a Lyapunov functional for it. Not an amplitude in the sense of a maximum displacement.
<i>Lyapunov functional</i>	Two different objects wear this name and they do opposite jobs. The <i>dissipation</i> functional Θ_d of Assumption 2.4 and the amplitude W are non- <i>increasing</i> along the motion and measure what it spends; they are what the counts of Section 5 rest on. The <i>dual pair</i> (Φ, Ψ) of Section 6 is non- <i>decreasing</i> along the motion, and its superlevel set $\{\Psi \geq 1\}$ encloses the reachable set; that pair is the certificate, and exhibiting one is a proof whether or not it is ever evaluated numerically. The first kind is in hand; constructing the second is objective 4 .
<i>averaging</i>	Always over one orbit, with a single phase (Lemma 4.6). Averaging over <i>several</i> phases, whose frequencies may cross, is a different problem, is not attempted anywhere below, and is the subject of objective 3 .
<i>the oscillator</i>	Always the motion of Eqs. (2.1) to (2.2) — the altitude about the glide. The <i>attitude</i> oscillator of Section 7 , three orders of magnitude faster, is named in full wherever it appears.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. The state is $x = (z, v, s)$, with $z \in \mathbb{R}$ an altitude, $v \in \mathbb{R}$ its rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. The system is

$$\dot{z} = v, \quad (2.1)$$

$$\dot{v} = c m_\theta^\varepsilon(z) - 1 - \kappa(s, c, \theta) v, \quad (2.2)$$

$$\dot{s} = F(s, c, \theta) + m_\theta^\varepsilon(z) \Gamma(s, c, \theta). \quad (2.3)$$

Here $c(\cdot)$ is the control and $\theta \in \Theta$ is a *selection* — a choice of admissible field from a bounded family, fixed along each trajectory and quantified over at the end ([Remark 2.3](#)). The small parameter $\varepsilon \in (0, 1]$ appears in one place only, inside the density: the *profile* m_θ is read on the layer's own scale,

$$m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon), \quad M_\theta^\varepsilon(z) = \varepsilon M_\theta(z/\varepsilon), \quad (2.4)$$

where M_θ is the antiderivative of m_θ with $M_\theta(0) = 0$, so that $(M_\theta^\varepsilon)' = m_\theta^\varepsilon$ and $M_\theta^\varepsilon(0) = 0$. The *potential* is

$$\Phi_{c,\theta}(z) = z - c M_\theta^\varepsilon(z), \quad \text{so} \quad \Phi'_{c,\theta}(z) = 1 - c m_\theta^\varepsilon(z), \quad (2.5)$$

and Eqs. (2.1) to (2.2) are a unit-mass particle in that potential with a weak damping. We write $\Phi_{c,\theta}$ without decoration, though it depends on ε : all of that dependence is the density's, and Lemma 3.4 is where it is unpacked. The *amplitude* is its energy,

$$W = \frac{1}{2}v^2 + \Phi_{c,\theta}(z). \quad (2.6)$$

The whole of Section 3 is an account of one object: the shape of this potential.

That ε sits inside the density and nowhere else is the whole of the scaling. The air is felt over a band of altitudes of thickness ε while the motion of interest is not thin: a large oscillation rises to an altitude of order one and returns, so a pass occupies a vanishing *fraction* of each oscillation without the oscillation itself becoming fast. That is the configuration in which a pass concentrates, and Lemmas 3.4 and 3.5 are the two statements that establish it — each of which also says what the other choice of scaling would have given instead.

Assumption 2.1 (H1: the wall). For each $\theta \in \Theta$ the profile $m_\theta \in C^1(\mathbb{R})$ satisfies $m_\theta > 0$ and $m'_\theta < 0$; $m_\theta(\xi) \rightarrow 0$ as $\xi \rightarrow +\infty$ with $\int_0^\infty m_\theta < \infty$; $m_\theta(\xi) \rightarrow \infty$ as $\xi \rightarrow -\infty$; and $1/c$ lies in the range of m_θ for every $c \in [c_{\min}, c_{\max}]$. We write $m_+ = \sup_\Theta m_\theta(0)$ and $M_\infty = \sup_\Theta \int_0^\infty m_\theta$, both finite and independent of ε .

Because m_θ decreases, m_+ bounds m_θ^ε at and above the section $z = 0$ for every ε , which is the only place a pointwise bound on the density is used below. Inside the band m_θ^ε is unbounded as $\varepsilon \rightarrow 0$ and no pointwise bound is taken: what is used there is its *impulse*, which Assumption 2.4 bounds.

The two one-sided conditions are the whole physical content: the air thins upward without bound on the integral, and thickens downward without limit. Nothing in what follows uses the shape of m_θ beyond Assumption 2.1, which is why θ may select it: an enclosure of an uncertain density is exactly a bounded family of admissible profiles.

Assumption 2.2 (H2: data). $[c_{\min}, c_{\max}] \subset (0, \infty)$, Θ and $S \subset \mathbb{R}^d$ are compact. The maps $F, \Gamma: S \times [c_{\min}, c_{\max}] \times \Theta \rightarrow \mathbb{R}^d$ and κ are continuous, bounded by $F_{\max}$, $\Gamma_{\max}$ and $\kappa_{\max}$, and Lipschitz in s uniformly in (c, θ) with constants κ_F, κ_Γ . Moreover $\kappa \geq \kappa_{\min} > 0$. The selection θ is a fixed element of Θ along each trajectory; only $c(\cdot)$ varies with time.

Remark 2.3 (Why the selection is fixed, and what that costs). The reachable set below is the union over $\theta \in \Theta$ of the sets reached at fixed θ , so the uncertainty is quantified over — it is simply not allowed to vary *during* a trajectory. The restriction is not cosmetic and the reason is worth stating, because it is exactly where a hidden hypothesis would otherwise sit. The potential Eq. (2.5) depends on θ , so if θ were a function of time the amplitude Eq. (2.6) would change as the potential moved beneath it, and Eq. (3.1) would acquire a term in $\partial_\theta M_\theta^\varepsilon(z)\dot{\theta}$ — which for a merely measurable $\theta(\cdot)$ is not a term one may bound, or in general even write. Every estimate that descends from the amplitude would go with it.

What does *not* depend on this is the budget. Assumption 2.4 asks for one functional that decreases for *all* (s, c, θ) simultaneously, so Eq. (5.1) holds for an arbitrary measurable $\theta(\cdot)$ as it stands; it is a common Lyapunov functional and is indifferent to how the selection moves. So the division is clean: the slow budget survives a time-varying selection and the fast amplitude law does not.

The cost is a genuine narrowing of the uncertainty class, and it narrows it in the direction that makes the enclosure *smaller*: the set reached by trajectories along which the atmosphere and the coefficients may vary arbitrarily in time is larger than the union taken here, so Theorem 4.2 bounds the latter and not the former. Enlarging the class to selections of bounded variation in altitude — which is what the physics supplies, the atmosphere being smooth in space and the mass lost monotonically — is objective 2, and it is stated there as a generalisation of the theorem rather than as a refinement of its constants.

Assumption 2.4 (H3: a Lyapunov functional). There is $\Theta_d \in C^1(\mathbb{R}^d)$ and a constant $c_0 > 0$ with

$$\nabla \Theta_d(s) \cdot F(s, c, \theta) \leq 0, \quad \nabla \Theta_d(s) \cdot \Gamma(s, c, \theta) \leq -c_0,$$

for all (s, c, θ) . Write $M = (\max_S \Theta_d - \min_S \Theta_d)/c_0$.

A Lyapunov functional for the slow motion, non-increasing along every trajectory for every control and every selection. The bound on its total variation over the compact corridor is what converts monotonicity

into a count: by Eq. (5.1) it bounds $\int m_\theta^\varepsilon d\tau$ uniformly in ε , and every count in Section 5 descends from that one inequality. In the application it is the specific mechanical energy, which falls at the drag power because neither the lift nor the Coriolis force does work. Section 7 notes a second monotone quantity the application supplies, carrying a different power of the density, and what it would cost to use.

Assumption 2.5 (H4: control). $c(\cdot)$ is absolutely continuous with values in $[c_{\min}, c_{\max}]$ and $|\dot{c}| \leq \dot{c}_{\max}$. (The sharper form, in which c is piecewise constant on intervals of length at least Δ , is Remark 3.2.)

Assumption 2.6 (H5: corridor). Every trajectory remains on $[0, T]$ in a compact corridor $\mathcal{X} = \{|z| \leq z_{\max}, |v| \leq v_{\max}, s \in S\}$, and $x(0) \in \mathcal{X}_0 \subseteq \mathcal{X}$. Write $W_{\max} = \frac{1}{2}v_{\max}^2 + z_{\max}$ for the largest amplitude it admits.

The corridor is a fixed box, independent of ε , and it is the scaling of Eq. (2.4) that permits this: the oscillations of interest have amplitude of order one, so bounding them costs nothing that shrinks. The one quantity that needs care is the antiderivative, since M_θ^ε is unbounded below as $\varepsilon \rightarrow 0$ at fixed $z < 0$. It is not unbounded where a trajectory can reach. Wherever $\Phi_{c,\theta}(z) \leq W_{\max}$ we have $-c M_\theta^\varepsilon(z) = \Phi_{c,\theta}(z) - z \leq W_{\max} + z_{\max}$, and above the section $|M_\theta^\varepsilon| \leq \varepsilon M_\infty$, so

$$M_* := M_\infty + \frac{W_{\max} + z_{\max}}{c_{\min}} \quad (2.7)$$

bounds $|M_\theta^\varepsilon|$ on the reachable part of the corridor, uniformly in ε .

Remark 2.7 (What is uniform, and what the hypotheses are not). Every constant named above is a bound over $S \times [c_{\min}, c_{\max}] \times \Theta$. No constant below depends on the particular control, the particular selection, or the particular initial point: that is the sense in which the result is uniform over the admissible family, and it is why an enclosure of an uncertain field is the natural setting rather than a complication. Note also what is absent. There is no transversality condition, no non-grazing condition, and no assumption that a pass is a single dip — the last is Lemma 2.8, a consequence of $m'_\theta < 0$. Every hypothesis is a condition on the data, checkable before any trajectory is computed.

Lemma 2.8 (One well, and one dip per period). *Under Assumption 2.1 the potential $\Phi_{c,\theta}$ is strictly convex, with*

$$\Phi_{c,\theta}''(z) = -\frac{c}{\varepsilon} m'_\theta(z/\varepsilon) > 0, \quad (2.8)$$

and attains its minimum at the unique $z_(c, \theta) = \varepsilon \xi_*$, where ξ_* is the unique point with $c m_\theta(\xi_*) = 1$: the glide sits inside the band, at height of order ε . Consequently every sublevel set $\{\Phi_{c,\theta} \leq W\}$ is a bounded interval $[z_-(W), z_+(W)]$, and at frozen (c, θ) every bounded motion of Eqs. (2.1) to (2.2) with $\kappa = 0$ is a single closed orbit — one closed curve in the (z, v) plane, traversed periodically — on which z attains exactly one minimum and one maximum.*

Proof. Differentiating Eq. (2.5) twice and using Eq. (2.4) gives Eq. (2.8), which is positive because $c > 0$, $\varepsilon > 0$ and $m'_\theta < 0$. A strictly convex C^2 function has at most one stationary point; $\Phi'_{c,\theta} = 1 - c m_\theta^\varepsilon$ changes sign exactly once because m_θ is strictly decreasing from $+\infty$ to 0 and $1/c$ is in its range, so z_* exists and is unique. Strict convexity makes each sublevel set an interval, bounded because $\Phi_{c,\theta} \rightarrow +\infty$ at both ends. At frozen (c, θ) and $\kappa = 0$ the quantity Eq. (2.6) is constant, so the orbit is the level curve $\{\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)\}$, which is a single closed curve meeting $v = 0$ exactly at the two endpoints of the interval. Those are the one minimum and the one maximum of z . $\square$

Lemma 2.8 is worth noting for what it is not: a hypothesis. That a pass is a single dip rather than a train of them is a property of the motion, and here it costs one sign on the density rather than a condition imposed on solutions.

3 Three regimes of one oscillator

Glide, phugoid and skip are not three systems but three energies in the single well of Lemma 2.8. This section says which is which, and the statement that organises everything is that ε sets no shape — only a clock.

Lemma 3.1 (The amplitude law). *Along every trajectory of Eqs. (2.1) to (2.3),*

$$\dot{W} = -\kappa(s, c, \theta) v^2 - M_\theta^\varepsilon(z) \dot{c}, \quad (3.1)$$

whatever the profile m_θ and whatever ε ; the only time-dependent datum the potential carries is c , which is why Eq. (3.1) closes (Remark 2.3). Consequently, with M_ the uniform bound Eq. (2.7),*

$$W(\tau) \leq W(0) + M_* \dot{c}_{\max} \tau \quad \text{for } \tau \in [0, T]. \quad (3.2)$$

Proof. Differentiate Eq. (2.6) along Eqs. (2.1) to (2.2):

$$\dot{W} = v\dot{v} + \Phi'_{c,\theta}(z) \dot{z} + \frac{\partial \Phi_{c,\theta}}{\partial c} \dot{c} = v(c m_\theta^\varepsilon - 1) - \kappa v^2 + (1 - c m_\theta^\varepsilon) v - M_\theta^\varepsilon(z) \dot{c},$$

where $\partial \Phi_{c,\theta}/\partial c = -M_\theta^\varepsilon(z)$ by Eq. (2.5). The two terms carrying $c m_\theta^\varepsilon$ cancel identically, which is Eq. (3.1); the cancellation uses only that the same $c m_\theta^\varepsilon$ appears in Eq. (2.2) and in $\Phi'_{c,\theta}$. Since $\kappa > 0$, Eq. (3.1) gives $\dot{W} \leq M_* \dot{c}_{\max}$, and Eq. (3.2) follows by integration. $\square$

So W is a Lyapunov functional for the fast motion, decaying except where the control pumps it, and Assumption 2.5 is what makes Eq. (3.2) an *a priori* bound rather than a hypothesis about solutions. Together with the functional of Assumption 2.4, one for the fast motion and one for the slow, these are the two monotone quantities on which every count below rests.

Remark 3.2 (The sharper form, from a finite guidance interval). If c is piecewise constant on intervals of length at least Δ — which is what a finite guidance interval supplies — then $\dot{c} = 0$ between switches and Eq. (3.1) reads $\dot{W} = -\kappa v^2$ exactly: W is non-increasing on each interval, and changes only by the jumps $|\Delta W| \leq M_* |\Delta c|$ at the at most T/Δ switching times. Equation (3.2) then holds with $M_* \dot{c}_{\max} \tau$ replaced by $M_* |\Delta c|_{\max} \lceil \tau/\Delta \rceil$, and LaSalle's argument applies interval by interval.

3.1 The orbit averages, and the time the oscillation may occupy

Recall from Section 1.2 that an *orbit* here is one closed curve in the (z, v) plane, never a trajectory about the planet. Two exact identities do most of the quantitative work, and both come from periodicity rather than from any property of the profile.

Lemma 3.3 (The mean density of an orbit). *Let $\langle \cdot \rangle$ denote the time average over one closed orbit of Eqs. (2.1) to (2.2) at frozen (c, θ) and $\kappa = 0$. Then*

$$\langle m_\theta^\varepsilon \rangle = \frac{1}{c}, \quad (3.3)$$

for every amplitude W , every profile satisfying Assumption 2.1 and every ε ; the impulse of such an orbit equals its period divided by c . Separately, and along the trajectory as it is rather than along any frozen orbit, Eq. (5.1) bounds the time the motion may spend where the air is appreciable: for every $\mu > 0$,

$$|\{\tau \in [0, T] : m_\theta^\varepsilon(z(\tau)) \geq \mu\}| \leq \frac{M}{\mu}. \quad (3.4)$$

Proof. Integrate Eq. (2.2) over one period at $\kappa = 0$. The left side is $[v] = 0$ because v is periodic, so $\int (c m_\theta^\varepsilon - 1) d\tau = 0$, that is $c \int m_\theta^\varepsilon d\tau = \int d\tau$, which is Eq. (3.3) on dividing by the period. The impulse of the orbit is $\int m_\theta^\varepsilon d\tau = \langle m_\theta^\varepsilon \rangle \times (\text{period}) = (\text{period})/c$. For Eq. (3.4), the set where $m_\theta^\varepsilon \geq \mu$ contributes at least μ times its measure to $\int_0^T m_\theta^\varepsilon d\tau$, which Eq. (5.1) bounds by M ; this uses no periodicity and holds for an arbitrary measurable selection. $\square$

Equation (3.3) is the reason the skeleton is the glide, and it is worth saying at once in what sense it is used. It is a statement about the frozen, undamped system, and the proof of Theorem 4.2 nowhere assumes the trajectory follows such an orbit; what it supplies is the *form* of the averaged field in Definition 4.1, and the error of using that field is estimated in Lemma 4.6 without any appeal to periodicity. With that said: Whatever the oscillation does, its time-averaged density equals the density at the equilibrium: the glide is not merely the bottom of the well but the *mean* of every oscillation about it. One consequence is used below: since $\langle m_\theta^\varepsilon \rangle$ does not depend on W , the averaged motion of the slow state does not either, so the amplitude is an output of the reduced description and not a driver of it.

3.2 What ε does, and what it does not

Lemma 3.4 (ε is a clock, and the amplitude it is read against). *Let $\xi = z/\varepsilon$ and $u = v/\sqrt{\varepsilon}$, and let $\Phi_{c,\theta}^0(\xi) = \xi - cM_\theta(\xi)$ be the ε -free potential of the profile. At frozen (c, θ) and $\kappa = 0$ the orbit through amplitude W is, in the (ξ, u) plane, the level set $\{\frac{1}{2}u^2 = \Omega - \Phi_{c,\theta}^0(\xi)\}$ at*

$$\Omega = W/\varepsilon, \quad (3.5)$$

which does not involve ε ; and $d\tau = \sqrt{\varepsilon} d\xi/u$, so every time measured along the orbit is $\sqrt{\varepsilon}$ times a quantity depending on (Ω, c, θ) alone. In particular the period and the time below the section $z = 0$ are

$$P = \sqrt{\varepsilon} P_0(\Omega, c, \theta), \quad P_{\text{dip}} = \sqrt{\varepsilon} D_0(\Omega, c, \theta). \quad (3.6)$$

Proof. Substituting $z = \varepsilon\xi$ in Eq. (2.4) gives $m_\theta^\varepsilon(z) = m_\theta(\xi)$ and $M_\theta^\varepsilon(z) = \varepsilon M_\theta(\xi)$, so $\Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi)$. Then $W - \Phi_{c,\theta}(z) = \varepsilon(\Omega - \Phi_{c,\theta}^0(\xi))$ and $\frac{1}{2}v^2 = \varepsilon \cdot \frac{1}{2}u^2$, which is the stated level set. Equation (2.1) gives $d\tau = dz/v = \varepsilon d\xi/(\sqrt{\varepsilon}u) = \sqrt{\varepsilon} d\xi/u$, so every elapsed time is $\sqrt{\varepsilon}$ times an integral over ξ of a function of (Ω, c, θ) . Taking it over the whole orbit and over its part with $\xi \leq 0$ gives Eq. (3.6). $\square$

So ε still sets no shape, only a clock — but the amplitude at which the clock is read is $\Omega = W/\varepsilon$, and that is what the scaling of Eq. (2.4) achieves. An oscillation of amplitude of order one is read at $\Omega \rightarrow \infty$, where the profile potential is dominated by the linear term and the orbit is a long ballistic arc with a brief rebound: P_0 grows like $2\sqrt{2\Omega}$ while D_0 grows only logarithmically, so by Eq. (3.6)

$$P \rightarrow 2\sqrt{2W}, \quad \frac{P_{\text{dip}}}{P} = \frac{D_0(W/\varepsilon)}{P_0(W/\varepsilon)} \rightarrow 0, \quad (3.7)$$

the period tending to a limit of order one and the dip fraction to zero. For the exponential profile $m_\theta = e^{-\xi}$ both are explicit, $D_0/P_0 = \log(4\Omega/c)/(2\Omega)(1 + o(1))$, so that

$$\frac{P_{\text{dip}}}{P} = \frac{\varepsilon}{2W} \log \frac{4W}{c\varepsilon} (1 + o(1)). \quad (3.8)$$

Had z been measured in layer thicknesses, Ω would have been W itself, D_0/P_0 would have been a fixed number at fixed amplitude, and nothing would have concentrated. This is the one place the choice of scaling does work, and Lemma 3.5 is the other.

Lemma 3.5 (The impulse of a pass is its entry speed). *Let a dip occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$, and write $I = \int_\alpha^\beta m_\theta^\varepsilon(z) d\tau$ for its impulse and $P_{\text{dip}} = \beta - \alpha$. Then, exactly,*

$$cI = (v(\beta) - v(\alpha)) + P_{\text{dip}} + \int_\alpha^\beta \kappa v d\tau. \quad (3.9)$$

At frozen (c, θ) the dip is elastic — $\Phi_{c,\theta}(0) = 0$, so Eq. (2.6) gives $\frac{1}{2}v^2 = W$ at both crossings and $v(\beta) = |v(\alpha)| = \sqrt{2W}$ — and then

$$\left| I - \frac{2\sqrt{2W}}{c} \right| \leq \frac{(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}}{c_{\min}}. \quad (3.10)$$

Proof. Integrate Eq. (2.2) over $[\alpha, \beta]$ and rearrange: that is Eq. (3.9), with no approximation of any kind. For Eq. (3.10), substitute $v(\beta) - v(\alpha) = 2\sqrt{2W}$ and bound the last two terms of Eq. (3.9) by $(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}$, then divide by $c \geq c_{\min}$. $\square$

Equation (3.10) is the statement that makes a pass a profile rather than a perturbation. By Eq. (3.7) the dip occupies a vanishing fraction of its orbit; by Eq. (3.10) it nonetheless delivers an impulse of order one, fixed by the speed at which the oscillation entered. The two together are concentration, stated with constants: the duration goes to zero and the effect does not. In the alternative scaling the same calculation returns $I = O(\sqrt{\varepsilon})$, and a pass leaves no trace.

3.3 The excursion decomposition, and the one threshold

The identities so far are either exact along the trajectory (Lemmas 3.1 and 3.5) or statements about the frozen, undamped system (Lemmas 3.3, 3.4 and 2.8). The proof must not confuse the two, so we fix the decomposition of $[0, T]$ by the trajectory's own crossings of the section and nothing else.

Definition 3.6 (Passes, arcs, and the fast part). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. A *pass* is a maximal subinterval of $[0, T]$ on which $z \leq 0$; an *arc* is a maximal subinterval on which $z \geq 0$. A pass is a *bubble* if the amplitude at the time it begins is at least W_0 , and an arc is *ballistic* if the amplitude at the time it begins is at least W_0 . The *fast part* is the union of the remaining passes and arcs — every interval on which the amplitude is below W_0 — and on it

$$|v| \leq v_1 := \sqrt{2(W_0 + |\Phi_{\min}|)}, \quad \Phi_{\min} = \min_z \Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi_*) = O(\varepsilon), \quad (3.11)$$

by Eq. (2.6) and Lemma 2.8. Nothing here refers to a periodic orbit: a pass is an interval between two crossings of a section, and that is all.

No orbit is ever asserted to be closed, and no trajectory is ever asserted to follow one. What the frozen system supplies is a *comparison*, used in exactly one place — to bound how long a pass can last — and that comparison is Lemma 3.7. Everything else descends from Eqs. (3.9) and (5.1) and from Eq. (3.11), all three of which hold along the trajectory as it is.

Lemma 3.7 (A pass is no longer than the frozen pass beneath it). *Let a pass occupy $[\alpha, \beta]$ with amplitude $W = W(\alpha) \geq W_0$ and control $c = c(\alpha)$, and set $\delta = (\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}$, which bounds the drift of the amplitude across the pass by Eq. (3.1). Then*

$$P_{\text{dip}} \leq \sqrt{\varepsilon} \sup \left\{ D_0(\Omega', c', \theta) : \Omega' \geq \frac{W-\delta}{\varepsilon}, |c' - c| \leq \dot{c}_{\max} P_{\text{dip}} \right\}. \quad (3.12)$$

Proof. On $[\alpha, \beta]$ the amplitude satisfies $W(\tau) \geq W - \delta$ by Eq. (3.1) integrated, and $|c(\tau) - c| \leq \dot{c}_{\max} P_{\text{dip}}$ by Assumption 2.5. By Eq. (2.6), $\frac{1}{2}v(\tau)^2 = W(\tau) - \Phi_{c(\tau),\theta}(z(\tau))$, so at each altitude the speed is at least the speed the frozen system would have at that altitude with amplitude $W - \delta$ and the least favourable admissible c' . Since $P_{\text{dip}} = 2 \int_{z_-}^0 dz/|v|$ and the integrand decreases when $|v|$ increases, replacing $|v|$ by that lower bound can only increase the integral, which gives Eq. (3.12) on rewriting the frozen integral through Eq. (3.6). The bound is implicit in P_{dip} through δ , and closes by continuity: δ is of the order of P_{dip} itself, so δ/ε is of the order of $\log(1/\varepsilon)$ while W/ε is of the order of $1/\varepsilon$, and the shift in Ω' is relatively negligible. $\square$

Equation (3.12) is the only inequality in the document that compares the trajectory with a frozen orbit, and it is worth saying why it is affordable here and would not be in general. A pass is short. Over an interval of length P_{dip} the amplitude and the control cannot move by more than $O(P_{\text{dip}})$, so the comparison is local and self-improving, and no adiabatic theorem over the whole horizon is needed. A construction that asked the trajectory to track a closed orbit over many periods would need one.

With Definition 3.6 fixed, one number sets both errors. Let

$$\vartheta_1 = \sqrt{\varepsilon} \sup \left\{ D_0(\Omega, c, \theta) : \Omega \geq \frac{W_0}{2\varepsilon}, c \in [c_{\min}, c_{\max}], \theta \in \Theta \right\}, \quad (3.13)$$

the longest pass any bubble can have, the factor 2 absorbing the shift of Eq. (3.12) for all small ε . There is no second threshold: the error of averaging the fast part is set by v_1 of Eq. (3.11) directly, as Lemma 4.6 shows, so W_0 alone governs both.

Lemma 3.8 (How small the threshold makes each error). *Under Assumptions 2.1 and 2.6 the supremum in Eq. (3.13) is finite, and as $\varepsilon \rightarrow 0$ at fixed W_0 ,*

$$\vartheta_1 \rightarrow 0, \quad v_1 \rightarrow \sqrt{2W_0}.$$

For the exponential profile the supremum is attained at the left endpoint once ε is small, and

$$\vartheta_1 = \frac{2\varepsilon}{\sqrt{W_0}} \log \frac{2W_0}{c\varepsilon} (1 + o(1)).$$

Proof. Finiteness is Eq. (3.6) with Ω confined by Eq. (3.5) and Assumption 2.6. By Eq. (3.7) and $P_0(\Omega) \rightarrow \infty$ we have $\sqrt{\varepsilon}D_0(\Omega) \rightarrow 0$ uniformly for $\Omega \geq W_0/(2\varepsilon)$, which gives the first limit; for the exponential profile D_0 is eventually decreasing in Ω by Eq. (3.8), so the supremum is its value at $\Omega = W_0/(2\varepsilon)$, which is $\sqrt{\varepsilon} \cdot 2 \log(4\Omega/c)/\sqrt{2\Omega}$ evaluated there. The second limit is Eq. (3.11) with $\Phi_{\min} = O(\varepsilon)$. No monotonicity of D_0 in Ω is assumed, which is why Eq. (3.13) is a supremum. $\square$

Remark 3.9 (What the threshold trades). The two errors move oppositely in W_0 , and choosing between them is Remark 4.4. Raising W_0 admits fewer bubbles and shortens each pass, so ϑ_1 falls; but it leaves more of the motion to be averaged at a larger speed excursion, so v_1 rises like $\sqrt{2W_0}$. Since ϑ_1 carries a factor $\varepsilon \log(1/\varepsilon)$ and v_1 does not, the best threshold shrinks with ε . The count of bubbles moves with it: by Eq. (3.10) each has impulse at least $2\sqrt{2W_0}/c_{\max}$, so Assumption 2.4 admits at most

$$N_* = \frac{c_{\max}M}{2\sqrt{2W_0}} \quad (3.14)$$

of them — a bound free of ε at fixed W_0 , obtained from an exact integral identity and a budget and from nothing else, and which grows as W_0 is lowered to buy a smaller residual. This is the profile count the classical argument obtains only by contradiction and without a constant.

3.4 The bubble

Definition 3.10 (Pass map). For a bubble, let (v^-, s^-) be its data where it last crosses $z = 0$ downward and (v^+, s^+) where it next crosses upward, computed from Eqs. (2.1) to (2.2) together with the *inner* slow equation $\dot{s} = m_\theta^\varepsilon(z) \Gamma(s, c, \theta)$ — that is, Eq. (2.3) with the drift F dropped, whose contribution over a dip is at most $F_{\max}P_{\text{dip}}$. The *pass map* is

$$J_\theta(v^-, s^-) = \{ (v^+, s^+) : \text{measurable } c(\cdot) \text{ on the dip } \},$$

at the trajectory's own θ ; we drop the subscript where it is fixed.

Lemma 3.11 (The bubble is nearly elastic, and bounded in closed form). *Let $v^- \leq -v_0 < 0$ with $v_0 = \sqrt{2W_0}$, which every bubble satisfies because $\frac{1}{2}(v^-)^2 = W \geq W_0$. Along every orbit of Definition 3.10:*

- (a) *at frozen (c, θ) the dip is elastic: since $\Phi_{c,\theta}(0) = 0$, Eq. (2.6) gives $\frac{1}{2}(v^\pm)^2 = W$ at both crossings, so $v^+ = |v^-|$ exactly;*
- (b) *in general W drifts across the dip by at most $(\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max})P_{\text{dip}}$ by Eq. (3.1), so*

$$|v^+ - |v^-|| \leq \frac{(\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max})P_{\text{dip}}}{v_0};$$

- (c) *the dip reaches no lower than the z_- with $\Phi_{c,\theta}(z_-) = W$, which exists and is unique by Lemma 2.8, and $|z_-| \rightarrow 0$ with ε ;*
- (d) *its impulse I obeys Eq. (3.10), and $|s^+ - s^-| \leq \Gamma_{\max}I$.*

Proof. (a) On the dip the orbit is the level set of Eq. (2.6), so $\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)$ throughout; at $z = 0$ the potential vanishes because $M_\theta^\varepsilon(0) = 0$, giving $\frac{1}{2}(v^\pm)^2 = W$ with the same W at both crossings. (b) Away from frozen data W is no longer constant, and Eq. (3.1) bounds its rate by $\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max}$ with M_* from Eq. (2.7); integrating over the dip and dividing by $|d(\frac{1}{2}v^2)/dv| = |v| \geq v_0$ gives the displayed bound. (c) is Lemma 2.8, and $z_- = \varepsilon\xi_-(\Omega)$ with $|\xi_-| = O(\log \Omega)$ by Lemma 3.4. (d) is Lemma 3.5 together with $s^+ - s^- = \int_{\text{dip}} m_\theta^\varepsilon \Gamma d\tau$. $\square$

Lemma 3.12 (The pass map, and how an error crosses it). *Let two entry data for the same (c, θ) differ by $(\delta v, \delta s)$, with both $|v^-| \geq v_0$. Then the exits differ by $(\delta v^+, \delta s^+)$ with*

$$|\delta v^+| = |\delta v|, \quad |\delta s^+| \leq e^{\kappa_\Gamma I} |\delta s| + \frac{2\Gamma_{\max}}{c_{\min}} |\delta v|, \quad (3.15)$$

where I is the impulse of the dip. Consequently J is Lipschitz in the Hausdorff distance, and — what the proof needs — the factor multiplying the slow error is exponential in that dip's own impulse.

Proof. Parametrise the dip by the delivered impulse $\mu = \int m_\theta^\varepsilon d\tau \in [0, I]$, in which $ds/d\mu = \Gamma(s, c, \theta)$, Lipschitz in s with constant κ_Γ and involving neither z nor v ; Gronwall over the horizon I gives the factor $e^{\kappa_\Gamma I}$ on δs . The speed is an isometry: by Lemma 3.11(a) $v^+ = |v^-|$ at frozen data, so $|\delta v^+| = |\delta v^-|$ with no amplification — there is no Gronwall to do on v , and in particular no factor $\varepsilon^{-1/2}$, which the inner equation for v carries but the exit condition removes. What remains is the dependence of the exit on δv through the horizon itself: by Eq. (3.10), $|\partial I/\partial v^-| \leq 2/c_{\min}$ up to $O(P_{\text{dip}})$, and $|\partial s^+/\partial I| \leq \Gamma_{\max}$, which is the second term of Eq. (3.15). Taking suprema over (c, θ) gives the Hausdorff statement. $\square$

The form of Eq. (3.15) is what matters downstream and is worth isolating. The slow error is amplified by $e^{\kappa_\Gamma I}$, a factor that tends to 1 as the dip's impulse tends to zero, so the product of these factors over any collection of bubbles is $e^{\kappa_\Gamma \Sigma I}$ and Assumption 2.4 bounds the exponent whatever the number of bubbles. The coupling from δv enters *additively* and not as a factor, and the speed error is not amplified at all; so the two cannot compound into a product over the passes. A bound stated as a single Lipschitz constant for J would lose exactly this, since such a constant exceeds 1 by a fixed amount and its power would diverge with the number of passes.

4 The limit system, and the theorem

Definition 4.1 (The limit system). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. The limit system $\mathcal{H}$ has state (z, v, s) on the arcs and (s) on the glide, with three transitions.

Arc (flow). $\dot{z} = v$, $\dot{v} = -1 - \kappa v$, $\dot{s} = F(s, c, \theta)$, while $z \geq 0$: the ballistic arc, on which the air is not felt.

Bubble (jump). At $z = 0$ with $v \leq -v_0$ the state is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

Glide (averaged flow). For $W < W_0$, $\dot{s} = F(s, c, \theta) + \frac{1}{c} \Gamma(s, c, \theta)$, by Eq. (3.3), and z is not carried.

$\mathcal{R}_\mathcal{H}(t) \subseteq \mathbb{R}^{2+d}$ is the set of states reached at time t from $\mathcal{X}_0$ over all admissible c , all $\theta \in \Theta$ and all selections from J , with the convention that a state on the glide is carried as $(0, 0, s)$ — the glide mode does not propagate (z, v) , and Eq. (4.1) says what that costs. We write $\mathcal{R}_\mathcal{H}(E) = \bigcup_{t \in E} \mathcal{R}_\mathcal{H}(t)$.

The convention needs its price stated, since $\mathcal{R}_\varepsilon(T)$ lives in $\mathbb{R}^{2+d}$ and the glide mode of $\mathcal{H}$ carries only s . Let $\Pi: \mathbb{R}^{2+d} \rightarrow \mathbb{R}^d$ be the projection on the slow coordinates and $\Lambda = \{(z, v) : \frac{1}{2}v^2 + \Phi_{c,\theta}(z) \leq W_0\}$ the set of fast states compatible with the fast part, which by Eq. (3.11) satisfies

$$\Lambda \subseteq \{|v| \leq v_1\} \times \{|z| \leq z_1\}, \quad z_1 = \sup\{|z| : \Phi_{c,\theta}(z) \leq W_0\}, \quad (4.1)$$

a set that shrinks with W_0 and with ε . A state the glide mode holds as s is to be read as the whole fibre $\Lambda \times \{s\}$ above it, and that is the lift Eq. (4.3) applies.

Note what the glide mode does *not* carry: by Eq. (3.3) the averaged slow field is independent of the amplitude, so no action coordinate is needed. The amplitude is an output of the reduced description. Note also that $\mathcal{H}$ is free of ε : the arcs are ballistic, the glide is averaged, and the only trace of the layer is the pass map, which Eq. (3.10) fixes to leading order in terms of the entry speed alone.

We measure the residual in $|(z, v, s)| = |z| + |v| + |s|$ and write $\mathbb{B}(\varrho)$ for the corresponding ball, $\oplus$ for the Minkowski sum. The constants are

$$C_f = F_{\max} + m_+ \Gamma_{\max}, \quad \kappa_0 = \kappa_F + \frac{\kappa_\Gamma}{c_{\min}}, \quad G = e^{\kappa_0 T + \kappa_\Gamma M}, \quad C_A = (\Gamma_{\max} + \kappa_\Gamma C_f T) K, \quad K = \frac{2}{c_{\min}} + \left(\frac{\dot{c}_{\max}}{c_{\min}^2} + \frac{\kappa_{\max}}{c_{\min}} \right) T, \quad (4.2)$$

with $m_+ = \sup_\Theta m_\theta(0)$ from Assumption 2.1 and M from Assumption 2.4. Every one of them is free of ε . That is not an accident of bookkeeping but the one discipline the proof keeps: wherever the density appears inside the band, where it is unbounded as $\varepsilon \rightarrow 0$, it is bounded by its *impulse* and never pointwise, and Assumption 2.4 bounds every impulse that occurs. The pointwise bound m_+ is used only above the section, where Assumption 2.1 makes it uniform.

Theorem 4.2 (Outer inclusion with an explicit residual). *Assume [Assumptions 2.1 to 2.6](#) and fix $W_0 \in (0, W_{\max}]$. Let ϑ_1 be the threshold [Eq. \(3.13\)](#), v_1 the speed bound [Eq. \(3.11\)](#), N_* the count [Eq. \(3.14\)](#), Λ the fibre [Eq. \(4.1\)](#), and A_ε the impulse felt on the ballistic arcs, for which $A_\varepsilon \leq \varepsilon M_\infty c_{\max} M / (2W_0)$. Then*

$$\mathcal{R}_\varepsilon(T) \subseteq \left[\underbrace{\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} + \Lambda \times \{0\} \right] \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad (4.3)$$

where the sum with $\Lambda \times \{0\}$ lifts the glide mode's slow states to the fast fibre they stand for, and

$$\varrho = G \left(\underbrace{F_{\max} N_* \vartheta_1}_{\text{the drift over the passes}} + \underbrace{C_A v_1}_{\text{averaging the fast part}} + \underbrace{\frac{2\Gamma_{\max}}{c_{\min}} N_* c_{\max} A_\varepsilon}_{\text{the air the arcs discard}} \right). \quad (4.4)$$

All constants are those of [Assumptions 2.1 to 2.6](#) through [Eq. \(4.2\)](#); none depends on the control, the selection, the initial point or ε . Taking the union over $\theta \in \Theta$ on both sides gives the statement for the whole admissible family.

Four remarks on the statement. The inclusion is over a short interval of times, of half-width the residual, because a pass of positive duration is compared with an instantaneous jump. The lift by Λ is not slack dressed as rigour: the glide mode genuinely does not know where in the fast fibre the motion sits, Λ is exactly the fibre the fast part can occupy, and it shrinks with W_0 alongside the residual. Each of the three terms of [Eq. \(4.4\)](#) is the error of exactly one approximation and is traced to one estimate below, so the theorem is modular: an improvement to any one of the three lemmas improves the residual without touching the others. And the theorem is stated at a *fixed* threshold, so ϱ is a number and not a rate; what the threshold does as $\varepsilon \rightarrow 0$ is [Remark 4.4](#).

Remark 4.3 (What the theorem is, and is not, as a decomposition). [Theorem 4.2](#) is not a profile decomposition and does not establish one. It compares a trajectory with a trajectory of $\mathcal{H}$, and the objects a decomposition would produce — profiles, their scales, their mutual orthogonality — appear nowhere in its statement or its proof. Two things are nevertheless true of it, and the distinction is the whole of the methodological claim. First, it is the quantitative estimate that any eventual decomposition must reproduce: a decomposition into a skeleton, the passes rescaled to their own widths, and a remainder would have to yield [Eq. \(4.4\)](#), and [Eq. \(3.14\)](#) is the profile count it would have to deliver. Second, and more to the point, it shows that the concentration structure can be reached *without* the decomposition machinery — by an exact integral identity and a budget. Whether the abstract machinery buys anything the elementary route does not is [objective 1](#), and it is a real question rather than a formality.

Remark 4.4 (The threshold at its best, and the rate that follows). By [Lemma 3.8](#) the first term of [Eq. \(4.4\)](#) falls like $\varepsilon \log(1/\varepsilon)/W_0$ and the second rises like $\sqrt{W_0}$, while the third falls like $\varepsilon/W_0^{3/2}$, so there is a best threshold and it shrinks with ε . Balancing the second against the third gives $W_0 \asymp \sqrt{\varepsilon}$ and

$$\varrho = O(\varepsilon^{1/4}), \quad (4.5)$$

at which threshold the first term is $O(\sqrt{\varepsilon} \log(1/\varepsilon))$ and is not binding — so the logarithm of [Eq. \(3.8\)](#) does not reach the rate. We record [Eq. \(4.5\)](#) for the order of magnitude and claim nothing about its sharpness: it is what balancing three crude bounds gives. Which of the three is the real obstruction is a quantitative question we would want settled early, and the third — the air felt on the arcs, which is an artefact of cutting a pass at $z = 0$ rather than at the edge of the band — is the one we would expect to improve first.

Remark 4.5 (At a fixed ε , and what the threshold then costs). Nothing above requires ε to be small: [Eq. \(4.4\)](#) holds at every ε and every W_0 , and at a fixed physical ε it is evaluated rather than estimated. What it then says is that the residual is small only if ϑ_1 and v_1 both are, and [Section 7](#) shows that for entry they are not: no threshold makes the passes brief and the averaged excursion small together. The remedy is not a new error term but a narrower description. The motion near the threshold, which neither treatment serves, may be *propagated exactly* — carrying the fast pair (z, v) and the unapproximated field [Eqs. \(2.1\) to \(2.3\)](#) — and this adds nothing to ϱ because nothing is approximated, and removes the lift Λ over those windows because (z, v) is then carried. What it costs is dimension and amplification: the reduced problem carries $2 + d$ coordinates instead of d there, and G acquires the factor $e^{\kappa_0 T_E}$ for the total duration T_E of those windows, which [Eq. \(3.4\)](#) bounds by $2c_{\max} M$ once the windows are taken to be the intervals on which $m_\theta^\varepsilon \geq 1/(2c_{\max})$. So the threshold trades sharpness against dimension, not sharpness against sharpness.

Lemma 4.6 (Averaging the fast part, with its constant). *Let s solve Eq. (2.3) on a subinterval of the fast part and let $\bar{s}$ solve $\dot{\bar{s}} = F(\bar{s}, c, \theta) + \frac{1}{c}\Gamma(\bar{s}, c, \theta)$ with the same initial value. Then*

$$|s - \bar{s}| \leq G C_A v_1$$

with G , C_A as in Eq. (4.2) and v_1 as in Eq. (3.11); the factor G is the amplification over the horizon, and it is the same G that Eq. (4.4) collects once in front of every term, so C_A carries none of it.

Proof. Put $e = s - \bar{s}$. Subtracting, $\dot{e} = [F(s) - F(\bar{s})] + m_\theta^\varepsilon[\Gamma(s) - \Gamma(\bar{s})] + (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})$, so $|\dot{e}| \leq (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon)|e| + |(m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})|$.

Everything turns on the primitive $R(\tau) = \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s})$, and it is bounded from the equation of motion, with no appeal to periodicity and none to a frozen orbit. By Eq. (2.2), $c m_\theta^\varepsilon - 1 = \dot{v} + \kappa v$, so $m_\theta^\varepsilon - \frac{1}{c} = (\dot{v} + \kappa v)/c$ and, integrating by parts in the first term,

$$R(\tau) = \left[\frac{v}{c} \right]_0^\tau + \int_0^\tau \frac{v \dot{c}}{c^2} d\tau + \int_0^\tau \frac{\kappa v}{c} d\tau, \quad (4.6)$$

an identity, exact for every control and every selection. On the fast part $|v| \leq v_1$ by Eq. (3.11), so $|R| \leq v_1(2/c_{\min} + (\dot{c}_{\max}/c_{\min}^2 + \kappa_{\max}/c_{\min})T) = v_1 K$. Note what has replaced what: the zero-mean cancellation that periodicity would have supplied is here the statement that v is *small* on the fast part, which is a property of the trajectory and not of an orbit.

Integrating the inhomogeneous term by parts against R ,

$$\left| \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})\Gamma(\bar{s}) \right| \leq |R| \Gamma_{\max} + |R| \int_0^\tau \left| \frac{d}{d\tau} \Gamma(\bar{s}) \right| \leq v_1 K (\Gamma_{\max} + \kappa_\Gamma C_f T),$$

using $|\dot{\bar{s}}| \leq C_f$. Gronwall's inequality in integrated form multiplies this by $\exp \int_0^T (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon) \leq e^{\kappa_F T + \kappa_\Gamma M} \leq G$, by Assumption 2.4 and not by a pointwise bound on the density, which gives the claim. The estimate does not accumulate over the successive excursions of the fast part: R is bounded over the whole horizon by Eq. (4.6), not excursion by excursion. $\square$

5 Proof of Theorem 4.2

Fix $\varepsilon \in (0, \varepsilon_1]$, an admissible trajectory $x_\varepsilon = (z, v, s)$ on $[0, T]$ with control $c(\cdot)$ and selection $\theta(\cdot)$, and $x_\varepsilon(0) \in \mathcal{X}_0$.

Step 1: the budget

By Assumption 2.4, $\frac{d}{d\tau} \Theta_d(s) = \nabla \Theta_d \cdot F + m_\theta^\varepsilon \nabla \Theta_d \cdot \Gamma \leq -c_0 m_\theta^\varepsilon$, so

$$\int_0^T m_\theta^\varepsilon(z) d\tau \leq M, \quad (5.1)$$

for every control, every selection and every ε .

Step 2: the excursions, and the three counts

By Definition 3.6 the interval $[0, T]$ is the disjoint union of passes and arcs, cut at the trajectory's own crossings of $z = 0$, each classified by the amplitude where it begins. No periodicity is used and no orbit is asserted to close. Three counts follow, and each descends from Eq. (5.1) together with one exact identity.

How many bubbles. By Eq. (3.10) a bubble entered at amplitude $W \geq W_0$ delivers impulse at least $2\sqrt{2}W_0/c_{\max}$, up to the $O(P_{\text{dip}})$ of Eq. (3.10) which is absorbed for small ε . Since the impulses of disjoint passes sum to at most M by Eq. (5.1), there are at most the N_* of Eq. (3.14). This is Eq. (3.9) and a budget, and nothing else.

How long the passes last in total. By Lemma 3.7 each bubble has $P_{\text{dip}} \leq \vartheta_1$, so

$$\sum_{\text{bubbles}} P_{\text{dip}} \leq N_* \vartheta_1. \quad (5.2)$$

How much air the ballistic arcs discard. On an arc $z \geq 0$, so $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon) \leq m_+$, and by Lemma 3.4 the impulse of a single arc is $\sqrt{\varepsilon} \int_0^{\xi^+} m_\theta(\xi) d\xi/u \leq \varepsilon 2M_\infty/\sqrt{2W_0}$ for an arc entered above the threshold. Summing over at most N_* of them,

$$A_\varepsilon := \sum_{\text{ballistic arcs}} \int m_\theta^\varepsilon d\tau \leq \frac{\varepsilon M_\infty c_{\max} M}{2W_0}. \quad (5.3)$$

Equations (5.2) and (5.3) are the two estimates that make the construction work. The number of passes grows as the threshold is lowered while each pass shortens like $\varepsilon \log(1/\varepsilon)$, and Eq. (5.2) says the product is controlled; the arcs each feel an impulse of order ε , and Eq. (5.3) says the total is of order ε/W_0 . Neither is a compactness argument and neither could be obtained from one.

Step 3: the three treatments exhaust the motion

By Definition 3.6 every point of $[0, T]$ lies in a bubble, in a ballistic arc, or in the fast part, and these are precisely the three modes of Definition 4.1. There is nothing to assume: the classification is by the sign of z and by whether the amplitude at the start of the excursion reaches W_0 , both of which are properties of the trajectory. What has to be quantified instead is how good each treatment is, and that is Lemma 3.8: the longest pass is ϑ_1 and the largest speed on the fast part is v_1 , both computable from the profile and the data, and both small for small ε at fixed W_0 .

Step 4: a bubble is a jump, up to the drift over its pass

Let a bubble occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$. By Definition 3.10 the exit data of the inner problem from $(v(\alpha), s(\alpha))$ differ from $(v(\beta), s(\beta))$ only through the one term dropped in Eq. (2.3), namely the drift F , whose integral over the pass is at most $F_{\max} P_{\text{dip}}$. Hence

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(F_{\max} P_{\text{dip}}), \quad (5.4)$$

and $z(\alpha) = z(\beta) = 0$ exactly, so the altitude contributes nothing. It is $F_{\max}$ and not C_f that appears here: the density term of Eq. (2.3) is retained by the inner problem, which is the only reason this step survives a density unbounded inside the band.

Step 5: the fast part is the averaged glide

This is Lemma 4.6, giving $G C_A v_1$ — the G being the horizon amplification that Eq. (4.4) collects in front of every term, counted here once and not again in Step 6. Note which averaging this is, and which it is not. It is not an averaging theorem over a periodic orbit: no orbit is assumed to close, the control varies, and the damping acts. Equation (4.6) replaces the zero-mean cancellation of the periodic argument by the observation that v is small wherever the motion is being averaged, which is Eq. (3.11) and a property of the trajectory. Averaging over *several* phases, whose frequencies may cross, is a different problem and is not attempted anywhere in this document; it is the subject of objective 3, where we propose not to attempt it at all.

Step 6: assembling, and why the amplification is bounded

Construct $x_{\mathcal{H}}$ by following $\mathcal{H}$ with the same c and the same θ , jumping at each bubble with the point of J nearest to x_ε 's exit, flowing ballistically on each ballistic arc, and flowing with the averaged field on the fast part. Write δ for the discrepancy and track its two components separately, which is what Eq. (3.15) is for.

The slow error is amplified in two places and in both the factor is exponential in an impulse. On the arcs and the fast part both motions obey fields Lipschitz with constant κ_0 , giving $e^{\kappa_0 T}$ over the horizon. Across a bubble, Eq. (3.15) gives the factor $e^{\kappa_\Gamma I}$ for a bubble of impulse I , so the product over all bubbles is

$$\prod_{\text{bubbles}} e^{\kappa_\Gamma I_k} = e^{\kappa_\Gamma \sum_k I_k} \leq e^{\kappa_\Gamma M}$$

by Eq. (5.1). Together the two are the G of Eq. (4.2). This is the second place the budget does work that no compactness argument could: the amplification compounds over every pass, however many there are, and is nonetheless bounded, because each factor is exponential in that pass's own impulse and the impulses are budget-limited. The step turns on the *form* of Eq. (3.15) and not merely on its size. Had the passage of an error across a bubble been recorded as a single Lipschitz constant for J , that constant would have exceeded 1 by a fixed amount and its product over the passes would have diverged as the threshold was lowered.

The speed error is not amplified at all, by Eq. (3.15): the bubble is an isometry in v . It is only *injected*, on the ballistic arcs, where x_ε feels the density that $x_\mathcal{H}$ does not, at a total cost of $c_{\max} A_\varepsilon$ by Eq. (5.3). It reaches the slow error through the additive coupling of Eq. (3.15), once per bubble, for a total of at most $(2\Gamma_{\max}/c_{\min}) N_* c_{\max} A_\varepsilon$ — the third term of Eq. (4.4), and the only one in which the number of passes appears.

Collecting Eqs. (5.2) and (5.4) for the bubbles, Step 5 for the fast part and Eq. (5.3) for the arcs,

$$|x_\varepsilon(T) - x_\mathcal{H}(\hat{T})| \leq G \left(F_{\max} \sum_{\text{bubbles}} P_{\text{dip}} + C_A v_1 + \frac{2\Gamma_{\max}}{c_{\min}} N_* c_{\max} A_\varepsilon \right),$$

which with Eq. (5.2) is the ϱ of Eq. (4.4), the comparison being made after both states are lifted to the fast fibre Λ of Eq. (4.1) wherever $x_\mathcal{H}$ is on the glide. The arrival time $\hat{T}$ differs from T because x_ε spends the passes where $x_\mathcal{H}$ spends no time, and by Eq. (5.2) that discrepancy is at most $N_* \vartheta_1$, which is at most $\varrho/(GF_{\max})$ and so at most ϱ whenever $GF_{\max} \geq 1$. Hence $\hat{T} \in [T - \varrho, T + \varrho]$ and Eq. (4.3) follows. $\square$

Remark 5.1 (Where each hypothesis was used). [Assumption 2.1](#) was used twice: in [Lemma 2.8](#), from which the shape of the well descends, and for the uniform bound m_+ on the density above the section. [Assumption 2.4](#) was used in Step 1, and everything quantitative descends from it: the pass count Eq. (3.14), the summed passes Eq. (5.2), the arc impulse Eq. (5.3), the time spent in appreciable air Eq. (3.4), the bounded amplification of Step 6, and the two places in [Lemma 4.6](#) where a pointwise bound on the density would have failed. [Assumption 2.5](#) was used twice, for the *a priori* amplitude bound Eq. (3.2) and for the $\dot{c}$ terms in [Lemma 3.7](#) and Eq. (4.6). [Assumption 2.2](#) supplies the constants and fixes the selection along a trajectory, without which Eq. (3.1) would not close ([Remark 2.3](#)); [Assumption 2.6](#) supplies the rest.

Two things are worth separating, because the proof would be wrong if they were run together. The statements about the frozen, undamped system — [Lemmas 3.3](#), [3.4](#) and [2.8](#) — are used to *define* the comparison quantities P_0, D_0 and the averaged field, and they enter the estimates in exactly one place, [Lemma 3.7](#), where the comparison is over a single short pass and is self-improving. The statements along the trajectory as it is — [Eqs. \(3.9\)](#), [\(3.11\)](#), [\(4.6\)](#) and [\(5.1\)](#) — carry everything else, and none of them assumes an orbit closes. There are five hypotheses and no sixth: that the three treatments cover the motion is [Definition 3.6](#), a definition. No compactness argument and no measure appears anywhere above.

6 The weak formulation

[Theorem 4.2](#) is a statement about trajectories, and its proof uses no measures. The weak formulation is nevertheless where the question belongs, for two reasons: it is what makes the solution set of a nonconvex family a tractable object at all, by rendering the dynamics a linear constraint; and it is the setting in which the loss of compactness that motivates the whole decomposition is visible. We give the definitions, so that what is open can be stated precisely.

For a trajectory of [Eqs. \(2.1\)](#) to [\(2.3\)](#) with control $c(\cdot)$ and selection $\theta(\cdot)$, write $\mathcal{U} = [c_{\min}, c_{\max}]$ for the control set of [Assumption 2.2](#). The *occupation measure* is the element $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ defined by

$$\int \varphi d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta(t)) dt \quad \text{for } \varphi \in C([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta),$$

together with the initial and terminal measures μ_0, μ_T carried by $x_\varepsilon(0)$ and $x_\varepsilon(T)$. The topology is the weak-* topology, that is, the topology of the dual of that space of continuous functions; the base is a compact metric space by [Assumption 2.2](#), so the topology is metrizable and bounded sets are relatively compact. In these terms the dynamics are a *linear* constraint: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) d\mu_T - \int \varphi(0, \cdot) d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (v, c m_\theta^\varepsilon - 1 - \kappa v, F + m_\theta^\varepsilon \Gamma) \right) d\mu_\varepsilon, \quad (6.1)$$

which is the Liouville equation of the family, and the reachable set at time T is the union of the supports of the terminal measures that satisfy it. This is the occupation-measure programme of Lasserre, Henrion and collaborators [\[5, 6, 7\]](#), and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$.

What [Theorem 4.2](#) does is to hand this machinery a different problem. After the theorem, the object to be enclosed is $\mathcal{R}_\mathcal{H}$: a problem in a few coordinates with a flow, one guard $z = 0$, a set-valued reset J , and a number of jumps bounded by the Lyapunov functional through [Eq. \(3.14\)](#). That is exactly the hybrid form for which the occupation-measure relaxation is known [\[8, 9\]](#): a measure on each guard carries the mass from one mode to the next, and the certificate Φ must be non-decreasing along the arc flow and across the reset. The singular limit has been discharged by [Section 5](#), which is elementary, and what is left on $\mathcal{H}$ is a problem in a few coordinates. Whether the limit is better taken in this topology, with the decomposition extracted from the measures, or at the level of trajectories as in [Section 5](#), is the open question of [Section 1.1](#), and the two routes are not known to be equivalent.

Remark 6.1 (What the reduction buys, and what now binds). The reduction is often put as a dimension count, and the count is real: the original problem carries thirteen coordinates and more, while $\mathcal{H}$ carries a few. But the dimension is not the binding constraint, because what is to be produced is a pair (Φ, Ψ) and not a mesh. A grid method for the Hamilton–Jacobi–Isaacs equation is exponential in the dimension and so has a hard ceiling near four or five states; a dual pair is *exhibited and verified*, and the dimension enters through the difficulty of finding one — and, if it is sought by a moment or sum-of-squares relaxation, through the size of the resulting programme — rather than as an exponential wall. Two consequences. The coordinate that [Section 7](#) would spend to use a second monotone quantity, and the fast pair that [Remark 4.5](#) carries through the orbits near the threshold, are costs rather than obstructions. And what does bind is the pass map: the condition on Φ at a jump is that it not decrease across it for *every* point of $J(v^-, s^-)$, so no certificate can be written down until J is enclosed. That is [objective 2](#), and it is not a question of dimension.

Two things are not done. [Theorem 4.2](#) compares trajectories and not measures, and the companion statement at the level of measures is [objective 4](#). And nothing above makes $\mathcal{R}_\mathcal{H}$ tight: an outer bound on it is an outer bound on $\mathcal{R}_\varepsilon(T)$ up to the residual, by [Eq. \(4.3\)](#), but how much is lost in passing from the one to the other is a separate question from how much is lost in the limit.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with the vehicle and its corridor entering only as data. Entry is not an arbitrary illustration, because it supplies the singular perturbation, the limit system and the concentrating events in one system. Nor is it already covered: the exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [\[10, 11\]](#), while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [\[12\]](#); and set-propagation methods return rigorous enclosures at ten to twenty states [\[13, 14, 15\]](#), but of one instance — one vehicle, one initial box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which is what a statement uniform over a family requires.

7.1 The correspondence

Measure altitude in planetary radii above the equilibrium glide, so that the atmospheric density is $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$ with the profile m_θ decreasing and read in scale heights, and let

$$\varepsilon = H_s/R_e$$

be the thin-atmosphere ratio, whose square root is the ratio of the oscillation period about the glide to the slow time $\sqrt{R_e/g}$. Then Eqs. (2.1) to (2.2) are the vertical motion of the vehicle: v is the rate of climb in those units, c is the vertical lift fraction $(L/D) \cos \sigma$ normalised so that $c m_\theta^\varepsilon = 1$ is the glide balance — lift against weight less centrifugal relief — and κ is the phugoid damping, which does not scale with ε . The slow state s collects the energy, the range and the heading, F carries the gravitational terms and Γ the drag. The selection θ chooses the density profile and the aerodynamic coefficients from an enclosure — one choice per flight, by Assumption 2.2 — while the control still ranges over a set at each instant, so that Eqs. (2.1) to (2.3) are a differential inclusion in the control over a family indexed by the uncertainty [16, 17], and Remark 2.7 is the statement that the bound holds over the whole enclosure. What is not covered is an atmosphere that varies during the flight itself, which is objective 2. Here the vehicle enters only through dimensionless ratios — c , κ , the budget constants — each quantified over a range for a class, never evaluated at a design.

The identification of the three regimes is the content of the model rather than an analogy. The minimum of the well is the equilibrium glide; the small oscillation about it is the phugoid, of period $\sqrt{\varepsilon}$ on the slow time scale and damping ratio of the same order, so that it persists for the whole entry — and it lives inside the band, at height of order ε , by Lemma 2.8; and a large oscillation is a skip, of period of order one, rebounding from the exponential wall in a time of order $\varepsilon \log(1/\varepsilon)$ and coasting on a long ballistic arc. That the dip is short and the arc long is Lemma 3.4 together with the asymmetry of $\Phi_{c,\theta}$, and that the rebound nevertheless delivers an impulse of order one is Lemma 3.5; nothing more enters.

Assumption 2.4 is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work. Taking $\Theta_d = \mathcal{E}$ gives $\nabla \Theta_d \cdot F = 0$ exactly, and $\nabla \Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower bounds on the drag coefficient and the speed, since the drag is strictly proportional to the density. The budget, the pass count and the bound on the time spent in the layer therefore hold for entry as they stand. The remaining hypotheses are conditions on the corridor. Normalising the gravity term to -1 in Eq. (2.2) requires $g - V^2/r$ bounded away from zero, so the corridor is bounded away from circular speed. The bank bounded away from ninety degrees gives $c \geq c_{\min} > 0$ in Assumption 2.2, and the last clause of Assumption 2.1 — that $1/c$ lie in the range of the profile — is the requirement that a glide altitude exist for every admissible bank. Assumption 2.2 also needs the corridor bounded away from the geographic poles, where the spherical chart is singular.

7.2 A second monotone quantity, and what it would cost

The application supplies a second quantity of the same kind as Assumption 2.4 but carrying a different power of the density: the integrated heat load, whose rate goes as the square root of the density where the drag power goes as the density itself. The two cut in opposite directions, and that is what makes the pair informative. For a fixed value of the first integral, concentrating the density *lowers* the second, so the first bounds how many deep excursions there can be while the second bounds how long the motion may linger at moderate density — which is a bound on the glide, and so on the number of fast orbits, that Assumption 2.4 alone does not give.

Using it is not free, and the cost is worth stating because it falls exactly on the dimension. To bound $\int \sqrt{m_\theta^\varepsilon} d\tau$ the slow equation must carry a term in $\sqrt{m_\theta^\varepsilon(z)}$ as well as one in $m_\theta^\varepsilon(z)$, since Γ does not depend on z and no condition on it alone can produce that power. And then the identity Eq. (3.3) has no counterpart: the orbit average of $\sqrt{m_\theta^\varepsilon}$ is a function of the amplitude, not a constant, so the averaged slow motion ceases to be amplitude-free and the glide mode of Definition 4.1 must carry W as a coordinate. That is one dimension in exchange for a bound on the glide, and which way the exchange is worth making is a question for the application rather than for the model.

7.3 Why the band is not empty

No hypothesis of [Theorem 4.2](#) fails for entry, save that the atmosphere is taken fixed along a flight ([Remark 2.3](#) and [objective 2](#)). What fails is that the two errors of [Eq. \(4.4\)](#) are not both small at the physical ε , and the reason is quantitative. A pass entered at a flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman–Loh variables [18, 19]. For the Earth $\sqrt{\varepsilon} \approx 0.034$, about two degrees, and entry angles are a few degrees: a pass is therefore neither short compared with the arc that follows it nor distinct in scale from the oscillation about the glide. In the language of [Eq. \(3.13\)](#), no threshold W_0 makes ϑ_1 and v_1 small together: the amplitude at which a pass becomes brief is the amplitude at which the excursion being averaged becomes large. So [Theorem 4.2](#) holds, but its residual is not small, and [Remark 4.5](#) rather than [Eq. \(4.5\)](#) is what applies. That costs no term in the residual; it costs dimension, over a total duration bounded by [Eq. \(3.4\)](#).

8 Plan of work

[Theorem 4.2](#) is Theorem C for one model, and it is discharged. Five obligations remain, and we state each with the obstruction it has to overcome. The first and fourth are the analysis; the second and third carry the model to the application; the fifth is a question about evaluating a bound rather than about proving one. The first is nearer than it was: the profile count [Eq. \(3.14\)](#) it needs is proved, and what is open is the decomposition that count would organise.

Objective 1. Theorem A: the profile decomposition, with a constant

We must prove the decomposition itself: that every sequence of trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The count is to be read as the classical theory reads it, above a fixed size: by [Eq. \(3.14\)](#) the passes of impulse at least $2\sqrt{2W_0}/c_{\max}$ number at most $c_{\max}M/(2\sqrt{2W_0})$, whatever ε . The classical route iterates the dichotomy of the concentration-compactness principle [1, 20] and, in the form that resolves each profile at its own scale, extracts a pairwise orthogonal family [2, 21]; orthogonality is produced by the construction rather than assumed, and profiles at distinct times are orthogonal here with nothing to prove, since the scale of a profile is the width of a pass and its centre the time at which the pass occurs.

What we expect to add is the constant. In that theory the number of profiles above a given size is finite because the conserved quantity is finite, and the argument is by contradiction; here [Assumption 2.4](#) supplies a Lyapunov functional whose total variation is bounded uniformly in ε , so the count is bounded *in advance* and explicitly, as [Eqs. \(3.14\)](#) and [\(5.2\)](#) already show. The objective is met by the decomposition stated with that count, with the remainder measured in the largest share of dissipation held by any window of fixed rescaled width — the norm in which a remainder of this kind is small, and in which it is not small in the conserved quantity [2, p. 136].

Objective 2. The statement at a fixed ε , and a smaller selection class

Theorem 4.2 holds for entry, but with a residual that Section 7 shows is not small at the physical ε . We must prove the statement Remark 4.5 describes: the orbits near the threshold carried by the unapproximated field, the reduced problem acquiring the fast pair over a total duration bounded by Eq. (3.4), and the residual gaining no new term. The enclosure of an intermediate orbit is a propagation over a bounded time rather than an asymptotic estimate, so what has to be controlled is its growth, not its order; the obligation is a bound on that growth which is explicit in the interval widths.

The same objective carries the generalisation the base theorem defers. By Assumption 2.2 the selection is fixed along each trajectory, and Remark 2.3 says why: a time-varying θ moves the potential beneath the amplitude, so Eq. (3.1) acquires a term that a merely measurable selection does not permit one to write. The budget Eq. (5.1) is indifferent to this, being a common Lyapunov functional, so what has to be rebuilt is the fast half of the argument and not the slow. We must therefore state a class of time-varying selections for which the amplitude law survives — bounded variation in altitude, monotonicity where it holds, which is what the physics supplies, the atmosphere being smooth in space and the mass lost monotonically — and carry it through Lemmas 3.1, 3.7, 3.11 and 3.12. This is a genuine enlargement of the enclosure and not a sharpening of its constants: the set reached when the atmosphere may vary during a flight is strictly larger than the union over fixed selections. The objective is met by the fixed- ε theorem, with the enclosure growth explicit in the interval widths, over a class of selections of stated regularity.

Objective 3. The remainder: motions that do not concentrate

A profile decomposition separates what concentrates from what does not, and the second part is the remainder. The entry problem has motions of that kind which Eqs. (2.1) to (2.3) does not carry: the attitude and the elastic modes oscillate for as long as there is air, and nothing about them concentrates, so in the decomposition they are not profiles at all. The wave-equation theory is careful to say that its remainder is not small in the conserved quantity, since it contains every motion that concentrating profiles cannot represent [2, p. 136], and the question is therefore in what sense the remainder *is* small here.

The classical answer is averaging [22], and for *several* phases it does not reach a bound. (Over a single orbit it does, and Lemma 4.6 uses it; the difficulty is entirely in the passage to more than one phase.) An averaging theorem gives an error to an order; with several phases it holds only away from resonance, while the attitude frequency grows with the square root of the dynamic pressure and so rises from zero at the start of every pass; and the results that carry a solution through a resonance hold outside a set of initial conditions of small measure, which an outer statement about a whole solution set cannot discard. We propose a Lyapunov functional in place of an averaging theorem — to bound what these motions *do* rather than to resolve what they are. They act on the translational motion only through the aerodynamic force, so it is the force that must be enclosed, and the enclosure may then be carried by the selection θ at no cost in dimension. Two facts are to be established. Where there is no air there is no torque, so the angular rates are bounded by conservation and the attitude has no aerodynamic effect in any case. Where there is air, the attitude damping rate is proportional to the product of density and speed, while the energy identity fixes the integral of density against the cube of speed, so the budget that bounds the dissipation also bounds the integrated damping from below — the same air does both. The objective is met by a bound on the aerodynamic force error, explicit in the dimensionless groups of the class and uniform over the corridor, with the elastic modes included or enclosed as a widened selection set.

Objective 4. The measure-level companion and the certificate

We must show that the occupation measures of Section 6 converge to measures satisfying the Liouville equation of the limit system, with the mass crossing each pass carried by a measure on the graph of J , uniformly over the admissible family, and exhibit the dual pair (Φ, Ψ) non-decreasing along the arc flow and across the reset. This is what makes Theorem 4.2 a statement about the limit of the family rather than a comparison of two trajectories, and it is where weak-* compactness and concentration-compactness in the form that resolves each profile at its own scale [1, 20] are expected to be needed — for the structure, with the constants coming from the functionals of Assumption 2.4 as in objective 1.

Objective 5. Validated evaluation of the certificate

The guarantee is to be verified in validated arithmetic rather than produced by a numerical solver, and this is the obligation set aside in Section 1.1. We must show that the closed-form constants, the enclosures of the passes they do not cover, and the reachable set of the limit system can be evaluated over the whole corridor with every rounding error accounted for. Two losses have to be managed and neither is removed by refining a step: repeated occurrences of a variable are treated as independent, and the image of a box under a flow is not a box, so re-enclosing it compounds [3, Chs. 4–5, 10]. What controls the second is the representation of the set [4, 15, 23]. The objective is met when every inequality the certificate asserts is discharged by a computation returning either a proof or an explicit failure to decide at the working precision.

8.1 Sequencing

Three phases rather than five parallel tracks. The first extends Theorem 4.2 to the planar entry problem, where the correspondence above is exact and Assumptions 2.1 and 2.4 are verified rather than assumed; the question of which of the three terms of Eq. (4.4) is the real obstruction, raised in Remark 4.4, belongs to it, and so does objective 1, since the profile count Eq. (3.14) is already in hand and what remains is to state the decomposition it bounds. The second is objective 2, in which the threshold of Eq. (3.13) and the selection class are the two dials on the same statement. The third is objective 3, and it should be reached early enough that a negative result is still recoverable; objective 4 runs alongside it, needing only that the limit system be settled, and objective 5 follows whatever the first phase settles, since it is a question about evaluating whichever bound that phase leaves. If the Lyapunov route of objective 3 does not close, the retreat is a corridor on which the frequency bands are ordered, with the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation; that weakens the residual without invalidating the decomposition.

References

- [1] P.-L. Lions. “The Concentration-Compactness Principle in the Calculus of Variations. The Locally Compact Case, Part 1”. In: *Annales de l’Institut Henri Poincaré, Analyse Non Linéaire* 1.2 (1984), pp. 109–145.
- [2] H. Bahouri and P. Gérard. “High Frequency Approximation of Solutions to Critical Nonlinear Wave Equations”. In: *American Journal of Mathematics* 121.1 (1999), pp. 131–175.
- [3] R. E. Moore, R. B. Kearfott, and M. J. Cloud. *Introduction to Interval Analysis*. Vol. 110. Society for Industrial and Applied Mathematics, 2009.
- [4] L. Jaulin et al. *Applied Interval Analysis: With Examples in Parameter and State Estimation, Robust Control and Robotics*. Springer-Verlag, 2001.
- [5] J.-B. Lasserre et al. “Nonlinear Optimal Control via Occupation Measures and LMI Relaxations”. In: *SIAM Journal on Control and Optimization* 47.4 (2008), pp. 1643–1666.
- [6] R. B. Vinter. “Convex Duality and Nonlinear Optimal Control”. In: *SIAM Journal on Control and Optimization* 31.2 (1993), pp. 518–538.
- [7] D. Henrion and M. Korda. “Convex Computation of the Region of Attraction of Polynomial Control Systems”. In: *IEEE Transactions on Automatic Control* 59.2 (2014), pp. 297–312.
- [8] V. Shia et al. “Convex Computation of the Reachable Set for Controlled Polynomial Hybrid Systems”. In: *53rd IEEE Conference on Decision and Control (CDC)*. 2014, pp. 1499–1506.
- [9] P. Zhao, S. Mohan, and R. Vasudevan. “Optimal Control of Polynomial Hybrid Systems via Convex Relaxations”. In: *IEEE Transactions on Automatic Control* 65.5 (2020), pp. 2062–2077.

- [10] S. Bansal et al. “Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances”. In: *56th IEEE Conference on Decision and Control (CDC)*. 2017, pp. 2242–2253.
- [11] I. M. Mitchell, A. M. Bayen, and C. J. Tomlin. “A Time-Dependent Hamilton-Jacobi Formulation of Reachable Sets for Continuous Dynamic Games”. In: *IEEE Transactions on Automatic Control* 50.7 (2005), pp. 947–957.
- [12] J. J. McNamara and P. P. Friedmann. “Aeroelastic and Aerothermoelastic Analysis in Hypersonic Flow: Past, Present, and Future”. In: *AIAA Journal* 49.6 (2011), pp. 1089–1122.
- [13] X. Chen, E. Ábrahám, and S. Sankaranarayanan. “Flow*: An Analyzer for Non-linear Hybrid Systems”. In: *Computer Aided Verification (CAV 2013)*. Ed. by N. Sharygina and H. Veith. Vol. 8044. Lecture Notes in Computer Science. Springer, 2013, pp. 258–263.
- [14] M. Althoff, G. Frehse, and A. Girard. “Set Propagation Techniques for Reachability Analysis”. In: *Annual Review of Control, Robotics, and Autonomous Systems* 4 (2021), pp. 369–395.
- [15] K. Makino and M. Berz. “Taylor Models and Other Validated Functional Inclusion Methods”. In: *International Journal of Pure and Applied Mathematics* 4.4 (2003), pp. 379–456.
- [16] A. F. Filippov. *Differential Equations with Discontinuous Righthand Sides*. Ed. by F. M. Arscott. Kluwer Academic Publishers, 1988.
- [17] J.-P. Aubin and A. Cellina. *Differential Inclusions: Set-Valued Maps and Viability Theory*. Vol. 264. Springer-Verlag, 1984.
- [18] D. R. Chapman. *An Approximate Analytical Method for Studying Entry into Planetary Atmospheres*. Tech. rep. Technical Note 4276. National Advisory Committee for Aeronautics, 1958.
- [19] W. H. T. Loh. *Re-entry and Planetary Entry Physics and Technology. Dynamics, Physics, Radiation, Heat Transfer and Ablation*. Ed. by W. H. T. Loh. Vol. 1. Springer-Verlag, 1968.
- [20] P.-L. Lions. “The Concentration-Compactness Principle in the Calculus of Variations. The Limit Case, Part 1”. In: *Revista Matemática Iberoamericana* 1.1 (1985), pp. 145–201.
- [21] P. Gérard. “Description du Défaut de Compacité de l’Injection de Sobolev”. In: *ESAIM: Control, Optimisation and Calculus of Variations* 3 (1998), pp. 213–233.
- [22] J. A. Sanders, F. Verhulst, and J. Murdock. *Averaging Methods in Nonlinear Dynamical Systems*. 2nd ed. Vol. 59. Springer, 2007.
- [23] N. S. Nedialkov, K. R. Jackson, and G. F. Corliss. “Validated Solutions of Initial Value Problems for Ordinary Differential Equations”. In: *Applied Mathematics and Computation* 105.1 (1999), pp. 21–68.